My 8th Grade Math Book In a Nutshell.

Chapter 1: Algebraic expressions

An algebraic expression is an expression that contains number, variables and operators.

Example:

1. 9x + 5
2. x – 2y + 3
3. $4x^3$ + xy + 2

Numbers: .., -1, 0, -2, …

Variable: a, b, c, …, z, α, β, …, Ω

Operator: +, -, ×, ÷, …

Terms are elements made up in an algebraic expression that is separated by + or - sign.

Example:

1. 9x + 5 → 9x and 5 are terms, there are 2 terms expression.
2. 4x + 2x -7 → 4x, 2x, and -7 are terms, there are 3 terms in this expression.

A polynomial is an algebraic expressions consisting of 2 or more terms. If there is only 1 term it is called monomial. Example:

1. 9x + 5 → Polynomial
2. 4x – 2y + 8 → Polynomial
3. 6y → Monomial

The degree of a polynomial is the largest sum of the exponents in a single term. In simple, words, pick a term, locate the variables then add the exponents, repeat for all term, pick the largest number of all. That's your degree for the polynomial.

Polynomial	Degree	Example
Constant polynomial	0	7
Linear polynomial	1	8x + 3
Quadratic polynomial	2	$3x^2$ - 1
Cubic polynomial	3	x^3 + 4x - 9
Quartic polynomial	4	xy^3- 9x + 1

Arithmetics with polynomials:

1. Adding 2 polynomials for example $(x^3+4x) + (4x^3 + x^2)$, you can just add directly resulting into $5x^3+x^2+4x$.
2. Subtracting 2 polynomials example $(x^3+4x) - (4x^3 + x^2)$, you need to invert the one being used as a subtractor in this case $(4x^3 + x^2) \rightarrow (-4x^3 - x^2)$. Then you can change the minus sign to an addition sign $(x^3+4x) + (-4x^3 - x^2) = -3x^3-x^2+4x$.
3. Multiplying 2 polynomial:

   $(x+4x)(4x - x^2)$
   $= (x+4x)(4x) + (x+4x)(-x^2)$
   $= (4x^2 + 16x^2) + (-x^3 + (-4x^3))$
   $= 20x^2 + (-5x^3)$
   $= -5x^3 + 20x^2$
4. Dividing 2 polynomials:

   $$\frac{8x+12}{4x^2-9}$$
   $$= \frac{4(2x+3)}{(2x+3)(2x-3)}$$
   $$= \frac{4}{2x-3}$$

   Or

   $$\frac{8x+12}{4}$$
   $= 2x + 3$

Factorization of polynomials

$x^2 + 4x + 3 = (x+1)(x+3)$

Factorising a polynomial means to express the polynomial by the product of its factor. In this case the factors are (x+1) and (x+3). The product (multiply) of both could be used to express the polynomial $x^2 + 4x + 3$ as they are the same value.

To factorise a polynomial you can use this method:

For every $ax^2 + bx + c$:

ax x	Factor of c = d c ÷ d = e	dx ax × e
ax^2	c	bx

In this method, make a 3x3 grid. On the first bottom column place ax^2 in your case. On the second bottom column place c in you case and on the third bottom column place bx in your case.

After all of that, the columns above ax^2 must be ax and x in your case. After that, the columns above c must be factors of c. On the first column, it must be 1 factor of c and the other column below needs to be another factor of c that is equal to c divided by the first factor you pick. After that, in the last row, bx's row, we need to cross multiply the ones we have inserted in the table. x is going to be multiplied with d, ax is going to be multiplied by e. After you have multiplied those, you will end up with 2 numbers. Add those numbers and see if it equals to bx. If it does, the factors are (ax + d)(x + e). If it doesn't try to pick another factor of c that you haven't tried out and do it until it equals to bx.

Expansion is the reverse of factorization:

(x+1)(x+3) = x^2 + 4x + 3

Additional formulas of factorization & expansion:

1. a^2-b^2 = (a+b)(a-b)
2. $(a+b)^2 = a^2 + 2ab + b^2$
3. $(a-b)^2 = a^2 - 2ab + b^2$
4. $a^3 + b^3 = (a+b)(a^2 - ab + b^2)$
5. $a^3 - b^3 = (a-b)(a^2 + ab + b^2)$

Binomial Theorem:

$$(a+b)^n = \binom{n}{0} a^n b^0 + \binom{n}{1} a^{n-1} b^1 + \binom{n}{2} a^{n-2} b^2 + \cdots + \binom{n}{n} a^0 b^n$$

Where:

1. $\binom{n}{r} = \frac{n!}{r!(n-r)!}$
2. $0! = 1$
3. a! = a × (a-1) × (a-2) × … × 1

Chapter 2: Statistics

Types of data:

1. Qualitative data are data that represent characteristics or description of something.
2. Quantitative data are data that can be measured or counted or calculated.

Example:

1. The most common name in town is Andy. → Qualitative data.
2. The most known cartoon show by the class is Upin & Ipin. → Qualitative data.
3. Average height of 6th graders. → Quantitative data.
4. The amount of customers on Monday. → Quantitative data.

Quantitative data has 2 types:

1. Discrete. It is counted. For example, customers in a shop.
2. Continuous. It is measured or calculated. For example, the length of a table.

Data can be collected in 2 types of way:

1. Directly from the source, called primary data.
2. Indirectly by collecting from someone or a place that got the data, called secondary data.

Data can be collected by observation, experiment, interview, survey etc.

In a data, the number of times a piece of information appear is called frequency. To determine frequencies we can use either counting or tallying. Tallying involves counting the frequency in groups of 5 rather than counting by counting manually and directly show the final result.

Example data:

Data of number of hours students study school materials outside school hour in a week:

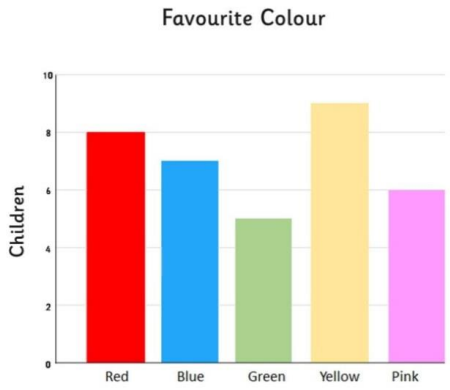
Number of hours	Tally	Frequency
20	𝍸	4
23	𝍸	3
27	𝍸	5
31	𝍸	8

Data can be represented in many ways:

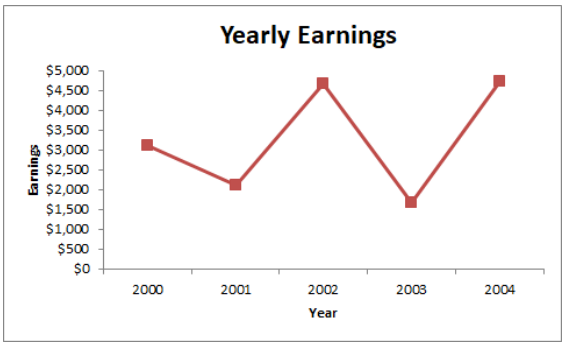
1. Pictogram | Source: Cuemath

Mode of transport	Number of students
Bus	☺☺☺☺☺☺☺
Car	☺☺☺☺
Walking	☺☺☺☺☺☺
Bycycle	☺☺☺
Key : ☺ Represents 4 children	

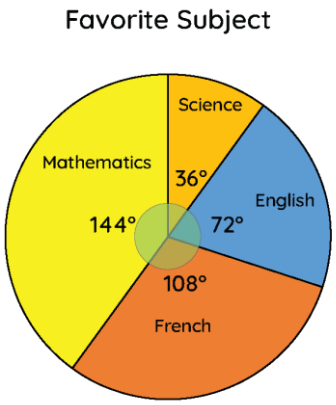
2. Bar Graph | Source: Twinkl



3. Line Graph | Source: Formplus



4. Pie chart | Source: Cuemath



Average of a data:

1. Mean
2. Mode
3. Median

Mean:

Mean can be calculated by the formula:

$$\frac{\Sigma x}{\Sigma f} = \frac{Total\ nominal\ of\ data}{Total\ freauency}$$

Mode:

The number that appears the most or the number that has the highest frequency.

Median:

To find the median you need to arrange the data in order, usually in an ascending way.

After that, you need to find the median position:

Median position = $\frac{\Sigma f+1}{2} = \frac{Total\ frequency + 1}{2}$

If the median position results in a positive whole number, for example 18. Then, the number in the 18th position after you have arranged the data will be the median.

If the median position results in a positive number with a decimal digit of 5 (e.g. 6.5 or 9.5 etc.) then remove the decimal. Then, find the digit in that position according to the median position without the decimal. Then, take also the digit after the digit you took before then add them both together and divide by 2 and that is your median. Example:

Data:

1, 3, 5, 7, 8, 9

Median position = $\frac{\Sigma f+1}{2}$

$= \frac{6+1}{2}$

= 3.5 → Take 3

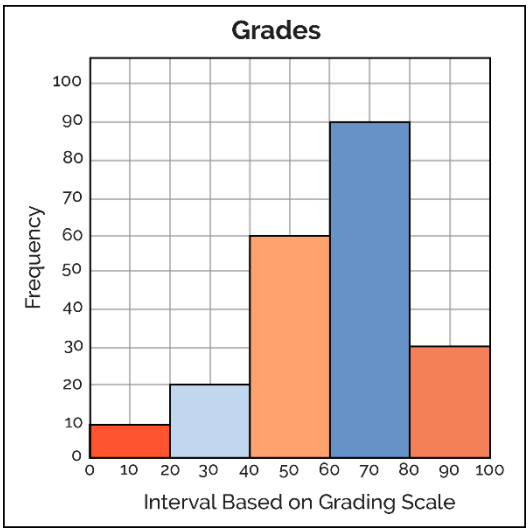
Number in 3rd position = 5

Number after it = 7

Median = $\frac{5+7}{2}$

= 6 → Median of that data is 6.

Histogram | Source: Data science discovery – University of Illinois



A histogram is like a bar graph but instead of single things having their own frequencies, they are grouped. In a bar graph, it could be 21 – 5 frequency, 26 – 8 frequency but in a histogram it will be 20-30 – 13 frequency. Same original data, different ways of interpreting it.

Dispersion of data:

Range tells us how wide a data is dispersed or spread. Range can be calculated by Largest value in data – smallest value in data.

Quartiles of a data:

Quartile can be used as a way to group data that has already been arranged into 4 parts.

In a data, there will always be 3 quartiles:

1. Q_1 = First quartile = Lower quartile
2. Q_2 = Second quartile = Median
3. Q_3 = Third quartile = Upper quartile

Interquartile range = $Q_3 - Q_1$

Quartile deviation (QD) = $\frac{Q_3 - Q_1}{2}$

Quartile position formula:

If ∑f even:

$$Q_k = \frac{\Sigma f \times k+2}{4}$$

If ∑f odd:

$$Q_k = \frac{\Sigma f \times k+k}{4}$$

Standard deviation = $\sigma = \sqrt{\frac{\Sigma(x-\bar{x})^2}{\Sigma f}}$

Variance = $Standard\ deviation^2 = \sigma^2$

Example:

Data:

1, 3, 7, 9, 11, 17

$$\bar{x} = \frac{\Sigma x}{\Sigma f}$$

$$= \frac{1+3+7+9+11+17}{6}$$

$$= 8$$

$$\sigma = \sqrt{\frac{\Sigma(x-\bar{x})^2}{\Sigma f}}$$

$$= \sqrt{\frac{(1-8)^2+(3-8)^2+(7-8)^2+(9-8)^2+(11-8)^2+(17-8)^2}{6}}$$

$$= \sqrt{\frac{49+25+1+1+9+81}{6}}$$

$$= \sqrt{\frac{166}{6}}$$

Chapter 3: Linear equation in 2 variables

An equation is considered linear when the exponent of all variable is 1. Note that the one thing that might be easy to miss is that variables that are in the denominator spot of a fraction is not considered a linear equation, why? Because if it is in the denominator spot of the fraction like $\frac{1}{x}$ it is not a linear equation because $\frac{1}{x} = x^{-1}$, the exponent is not 1.

Example of a linear equation in 2 variables:

1. 6x – 2y = 7
2. p – 2q
3. 6m + 4n = 90

Simultaneous linear equation in 2 variables:

ax + by = e

cx + dy = f

Where a, b, c, d are coefficients, x and y are variables and e and f are constants.

From the form above, we can tell how many solutions are there for 2 of the simultaneous linear equation:

$$If\ \frac{a}{c} \neq \frac{b}{d}\ then\ 1\ solution$$

$$If\ \frac{a}{c} = \frac{b}{d} \neq \frac{e}{f}\ \ then\ no\ solutions$$

$$If\ \frac{a}{c} = \frac{b}{d} = \frac{e}{f}\ then \infty\ solutions$$

Solving a simultaneous linear equation:

There is always 2 common ways of doing it, elimination and substitution.

Elimination method:

5x – 6y = 7 When x9 → 45x – 54y = 63

9x + 7y = 10 When x5 → 45x + 35y = 50

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-89y = 13

$-y = \frac{13}{89}$

$y = -\frac{13}{89}$

$5x - 6y = 7$ When x7 → $35x - 42y = 49$

$9x + 7y = 10$ When x6 → $54x + 42y = 60$

---------- +

$89x = 109$

$x = \frac{109}{89}$

Substitution method:

$5x - 6y = 7$

$9x + 7y = 10$

$5x - 6y = 7$

$5x = 7 + 6y$

$x = \frac{7+6y}{5}$

$9x + 7y = 10$

$9\left(\frac{7+6y}{5}\right) + 7y = 10$

$\frac{63+54y}{5} + 7y = 10$ x5

$63 + 54y + 35y = 50$

$89y = -13$

$y = -\frac{13}{89}$

$x = \frac{7+6y}{5}$

$x = \frac{7+6\left(-\frac{13}{89}\right)}{5}$

$x = \frac{\frac{623}{89} + -\frac{78}{89}}{5}$

$x = \frac{545}{89} \times \frac{1}{5}$

$x = \frac{109}{89}$

Both ways are correct, just different interpretations.

Chapter 4: Relations and Functions